

ProvBench-RAG: Provenance-Sensitive Retrieval Under Near-Duplicate Source Confusion

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Abstract

Retrieval-augmented generation (RAG) systems are often evaluated on whether an answer can be supported by *some* retrieved passage. Deployed assistants, however, must routinely choose among near-duplicate sources—mirrors, stale snapshots, scraped copies, and jurisdiction-specific variants—that share topical overlap but differ in authority, validity, and attribution. We study this *instance-level provenance* problem under lexical collision: many candidates look equally relevant, yet only one (or none) should be cited.

We present ProvBench-RAG, a benchmark specification and reproducible dataset format with clustered documents, gold preferred sources, minimal evidence sets, temporal and jurisdiction constraints, and abstention when authoritative sources disagree. We further introduce PACER (Provenance-Aware Cluster Evidence Reranking), a transparent reranker that fuses min-max normalized BM25 scores with declarative provenance signals (authority, freshness, originality/copy labels, jurisdiction alignment) over a fixed candidate pool.

On a controlled synthetic split of 1,500 queries with explicit lexical decoys, BM25 achieves preferred-source accuracy at rank 1 (PSA@1) of 0 while PACER reaches 0.75 (McNemar $p < 10^{-200}$, $n = 1,200$), with composite ProvScore rising from 0.469 to 0.703 and nDCG@10 from 0.553 to 0.926. Stratified analysis shows gains across freshness, authority, and jurisdiction families. On 140 English Wikipedia revision clusters, PACER improves PSA@1 from 0.014 to 0.057 ($p = 0.031$), though absolute accuracy remains low. A pooled heterogeneous index (1,640 queries) shows recall collapse under a single lexical index (0.435) and partial recovery with provenance reranking (0.782). All results regenerate from the released artifact.